Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 44: Approved Solutions

Growth of Functions

Covers Chartrand & Zhang, Section 6.2

Status of this set

Ungraded practice. Solutions released with it. Every big-O claim needs its witnesses.

Part I: How to Practice This Material

Proving $f = O(g)$ is exhibiting a witness

Name a constant C and an integer k, then verify $f(n) \le C\,g(n)$ for all $n \ge k$. That is the whole proof — an existence claim, as in Lecture 26.
A useful habit: bound each term of f separately by a multiple of g, then add the constants. For $5n + 7$ against n^2: $5n \le 5n^2$ and $7 \le 7n^2$ for $n \ge 1$, so $C = 12$ works.

Disproving $f = O(g)$

Show the ratio $f(n)/g(n)$ grows without bound. If $f(n) \le C\,g(n)$ held for all large n, that ratio would stay below the fixed number C.

Three mistakes to expect

1. **Asserting $f = O(g)$ with no C and k.** The definition asks for them.
2. **Treating big-O as symmetric.** $f = O(g)$ does not give $g = O(f)$.
3. **Proving only one inequality for big-theta.** It needs both.

Part II: Practice Problems

Section A — Big-O with witnesses

1. Prove each, giving explicit C and k.

(a) $3n + 5 = O(n)$

(b) $n^2 + 3n + 2 = O(n^2)$

(c) $4n^3 + 2n = O(n^3)$

Approved solution

(a) For $n \geq 1$: $3n \leq 3n$ and $5 \leq 5n$, so $3n + 5 \leq 8n$. Take $C = 8$, $k = 1$.
(b) For $n \geq 1$: $n^2 \leq n^2$, $3n \leq 3n^2$, $2 \leq 2n^2$, so the sum is at most $6n^2$. Take $C = 6$, $k = 1$.
(c) For $n \geq 1$: $4n^3 + 2n \leq 4n^3 + 2n^3 = 6n^3$. Take $C = 6$, $k = 1$.

The pattern is the same every time: bound each term by a multiple of g, then add the coefficients. Smaller constants also work; any valid pair earns full credit.

2. Show that $n^2 \neq O(n)$.

Approved solution

Suppose $n^2 \leq Cn$ for all $n \geq k$, for some constant C. Dividing by n (positive) gives $n \leq C$ for all $n \geq k$ — but n grows without bound while C is fixed. Taking $n = \max(k, C + 1)$ contradicts the inequality. So no such C exists. ■

This is proof by contradiction — assume the witnesses exist, derive an absurdity. Lecture 27's method, applied to a definition that asserts existence.

Section B — Which are $O(n)$?

3. For each function, decide whether $f(n) = O(n)$, with a reason.

(a) $f(n) = 100$

(b) $f(n) = 2n + 5$

(c) $f(n) = \lfloor n/2 \rfloor$

(d) $f(n) = n^2 + 3n + 2$

Approved solution

(a) Yes. $100 \leq 100n$ for $n \geq 1$. A constant function is $O(n)$ — indeed $O(1)$.
(b) Yes. $2n + 5 \leq 7n$ for $n \geq 1$.
(c) Yes. $\lfloor n/2 \rfloor \leq n/2 \leq n$, so $C = 1$, $k = 1$.
(d) No. The ratio $(n^2 + 3n + 2)/n$ exceeds n, which is unbounded.

(a) is worth a remark: every constant function is $O(n)$, but the sharper statement is $O(1)$. Big-O gives an *upper* bound, and a loose upper bound is still true.

Section C — Big-theta

4. Show that $f(n) = 3n^2 + 2n$ and $g(n) = n^2$ satisfy $f = \Theta(g)$.

Approved solution

Upper. For $n \geq 1$: $3n^2 + 2n \leq 3n^2 + 2n^2 = 5n^2$, so $f = O(g)$ with $C_2 = 5$.
Lower. For $n \geq 1$: $3n^2 + 2n \geq 3n^2 \geq n^2$, so $C_1 = 1$.
Hence $1 \cdot g(n) \leq f(n) \leq 5 \cdot g(n)$ for $n \geq 1$, and $f = \Theta(g)$ with $k = 1$. ■

By Theorem 6.20 it would also suffice to show $f = O(g)$ and $g = O(f)$ separately; the two routes are the same work.

5. For each f, give the smallest nonnegative integer k with $f(n) = O(n^k)$.

(a) $f(n) = n^2 + n \log n$

(b) $f(n) = \dfrac{2n^3 + n + 1}{n^2 + 1}$

(c) $f(n) = (n^2 + 1)(n^3 + 1)$

Approved solution

(a) $k = 2$. Since $\log n \le n$, we have $n \log n \le n^2$, so $f(n) \le 2n^2$. And $f(n) \ge n^2$, so $k = 1$ fails.
(b) $k = 1$. For large n the quotient behaves like $2n^3/n^2 = 2n$. Formally $f(n) \le (2n^3+n+1)/n^2 \le 4n$ for $n \ge 1$.
(c) $k = 5$. Expanding gives $n^5 + n^3 + n^2 + 1 \le 4n^5$, and the leading term is n^5, so no smaller k works.

Part (b) is the one to watch. A quotient's degree is the difference of the degrees — $3 - 2 = 1$ — not the numerator's. Students who answer 3 have read only the top.

Section D — Writing

6. In four to six complete sentences, explain why $f = O(g)$ does not imply $g = O(f)$, and what additional condition makes the two equivalent.

Approved solution

Model answer. The statement $f = O(g)$ says only that f is eventually bounded above by a constant multiple of g, which leaves open the possibility that g grows much faster. For instance $2n^2 = O(n^3)$, but n^3 is not $O(2n^2)$, because the ratio $n^3/2n^2 = n/2$ grows without bound and so cannot stay under any fixed constant. Big-O is therefore a one-directional relation, more like $\le$ than like $=$, despite the notation. The condition that makes the relation symmetric is big-theta: by Theorem 6.20, $f = \Theta(g)$ holds exactly when $f = O(g)$ and $g = O(f)$ both hold, and then the two functions grow at the same rate.

Full credit requires a concrete pair and the observation that big-O behaves like an inequality. The notation $f = O(g)$ is misleading on this point and students should say so.

Instructor notes

- **Insist on C and k throughout.** A big-O claim without witnesses is an assertion, and the definition is explicitly existential.
- **Problem 2** is a contradiction proof in disguise. Point that out — it shows Chapter 3 is still in use.
- **Problem 5(b)** catches students who read only the numerator's degree.
- **Problem 6** addresses the notation itself: $f = O(g)$ uses an equals sign for a relation that is not symmetric. Worth saying aloud.
- Reserved for Homework 15: Section 6.2 even-numbered exercises.